BSc Project Log Book

Student Name: Hector Gittos

Project Name: NeuroView AI: An Explainable Deep Learning Platform for Clinical-Grade Brain Tumour Diagnosis and Radiologist Collaboration

Internal Supervisor: David Wakeling

External Supervisor: (if any)

Log of Activities

Date	Key activities	Outcomes	Comments/Action plans
Start October	Meeting the group, socialising	Discussed the organisation of supervisor groups	Reflection about chosen topic
11/10/24	Read through the literature review specs	Started elaborating a plan to stay organised/avoid last minutes	Began research on the chosen topic, identifying the problem
18/10/24	Shared what we have learnt so far on our chosen topics	Started understanding the structure of the literature review, what is expected of us	Reading about our chosen topic, extensive research on deep learning and CNNs and transfer learning as a better solution for generalisability. Looking for datasets available. The recommended one is from figshare although this doesn't contain no-tumour cases.
24/10/24	One month until submission of Literature review — clock is ticking	David motivated us to really start writing as soon as possible to not fall behind and to use reading week to get as much done as possible	Extensive research of deep learning models for the task, analysing different techniques, spent the reading week working balancing courseworks and literature review. Found my dataset. Worked out my goal: a radiologist platform with transfer learning models integrated
08/11/24	Debrief of progression after reading week	Satisfactory progress, but more work to anticipate any issues that may arise	Even more progress needs to be made for the literature review, the literature section needs to be clearer and more approaches need to be explored. The UI for the interface needs cleaning up, fix frontend errors for display in project plan

15/11/24	Debrief of progression	Good progress, discussed about the structure of the project plan further	Need to make my project plan clearer and clean up my references, literature part is looking good
22/11/24	Day before submission, discussed who has submitted and who might use their extention	Literature Review had been finalised, David gave me a few last tips on the structure	Submitted mine later, wished to hear any feedback or recommendations before submitting
29/11/24	Debrief about the submission, confidence in our work	Happy with my work although I realised a few issues with my structure / section organisation with a main section Literature Review followed by another main section CNN which should have been a subsection	Understood my mistake and proceeded with reading more about transfer learning and different models.
05/12/24	Debrief about progress, how we are feeling so far, what could we do better	Progress is good, starting to have an idea of which transfer learning models to use : EfficientNet, DenseNet, ResNet etc	Continuing my research to work out which model to use and adapt to the Kaggle dataset
13/12/24	Last supervision meeting before Christmas holidays	Plan of action, use these couple of weeks to spend as much time as you can to start planning out your diss before courseworks start immerging during 2nd term	Carried on research and encountered Grad-CAM, a form of explainable AI that highlights regions of interest for model prediction. Wish to add that to my radiology interface.
13/01/25	Back from Christmas holidays, debriefed progress	Had a first draft structure lined up, what elements I wanted to have. I explained that I chose ResNet50 transfer learning approach and got a model with over 99% accuracy. Showed David my GradCAM working although it wasn't highlighting the relevant regions	Had issues with the GradCAM, the gradients and tensors. Session clearing wasn't working properly too. Continued with research about Explainable AI and thought about getting in touch woth a radiologist for my research — began searching
20/01/25	Debrief of progress	The GradCAM was working better, confidence scores were more esthetic.	Continued my search for radiologists to support my work, while looking for more explainability AI techniques. The project goal is becoming clearer.
27/01/25	Could not attend (attempted to attend via Microsoft teams but communication wasn't good)	Using Mixture of Experts for my work was suggested as I started making a second model to compete with the first one (MobileNetV3). I told David via Teams message that I managed	Got in touch more with Dr. Gaillandre, explained my research. Continued coding the platform. Analysis View was looking good, though about extending to patients, access to latest research studies (via API)

		to get in touch with a radiologist. Great outcome	
03/02/25	Debrief of progress, discussed submissions to come: poster demonstration, Final Report, Code/logbook and supervisor meeting log	Gained an understanding about these and started forward planning for what is to come and how to balance courseworks with time for dissertation	Carried on coding the interface. ResNet50 had 99.25% accuracy and MobileNetV3 over 98%.
10/02/25	Exchange about each other's progress	Good progress overall, the analysis panel works well, the patients view is being designed. Realised about the duplication issue in the dataset, discussed with David.	Continued developing a lot, fixing the data leakage issue that wasn't mentioned in the dataset description. New idea : AI chatbot for giving additional insights about the prediction
17/02/25	Debriefed about my chatbot idea	David pointed out the ethical concerns of adding MRI scan and sending it to OpenAI.	Removed the input image, assistant is now an assistive tool and does not have an effect on the diagnosis apart from providing medical information.
24/02/25	Discussion about any new ideas implemented	The AI assistant has been fixed, works well. Patient section is looking more aesthetic, although studies section needs to be implemented	Started Taking notes for my final report, added references on Notion, linking studies together. More coding for the interface. Patients and studies and nearly finished.
03/03/25	Debrief about progress, discussed submission of enterprise computing CA	Patient section and studies are done, workflow and navigation from patients to analysis works	Adding more references to Notion, from general information about radiologists, XAI to more specific model architecture decision choices. Wrote down th success criteria to evaluate NeuroView AI. Google Meet with Dr. Gaillandre to start testing the radiologist interface, great outcome.
10/03/25	Debrief about progress	More work about research, a first introduction draft has been written, background section halfway though	Added notes taken from Google Meet with Dr. Gaillandre in a logical flow, which helped me build my success criteria followed by results section for my dissertation.
17/03/25	Debrief about progress, discussed the submission of the poster. Changed	Introduction and background have a rough draft. Design and implementation for both AI	In touch with Dr.Gaillandre, sending him updates about the radiology platform. Continuing to

	from A0 to A1. Discussed the report progress	models is being made (preprocessing, describing model architecture, data augmentation etc)	write the report, reflecting on what to add in the discussion section.
24/03/25	Final dissertation meeting, last push during April before submission of the Report + code + logbook. Submission of poster in two days	Happy with the radiology interface, balancing coursework with continuing writing the report. Design and Implementation in good shape for both AI models. The poster is nearly done.	Finalised the poster with more extensions in mind to add in April such as the Radiologist's approval of diagnosis or reject to give full power to the medical expert. Worked on the dissertation throughout entire of April.